

Banker's Algorithm - C Program

C Program

```
#include <stdio.h>

int main() {
    int n, m, i, j, k;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter number of resources: ");
    scanf("%d", &m);

    int alloc[n][m], max[n][m], need[n][m];
    int avail[m], finish[n], safe[n];

    printf("Enter Allocation Matrix:\n");
    for(i=0; i<n; i++) {
        for(j=0; j<m; j++) {
            scanf("%d", &alloc[i][j]);
        }
    }

    printf("Enter Max Matrix:\n");
    for(i=0; i<n; i++) {
        for(j=0; j<m; j++) {
            scanf("%d", &max[i][j]);
        }
    }

    printf("Enter Available Resources:\n");
    for(i=0; i<m; i++) {
        scanf("%d", &avail[i]);
    }

    for(i=0; i<n; i++) {
        for(j=0; j<m; j++) {
            need[i][j] = max[i][j] - alloc[i][j];
        }
    }

    for(i=0; i<n; i++) {
        finish[i] = 0;
    }

    int count = 0;

    while(count < n) {
        for(i=0; i<n; i++) {
            if(finish[i] == 0) {
                int flag = 0;

                for(j=0; j<m; j++) {
                    if(need[i][j] > avail[j]) {
                        flag = 1;
                        break;
                    }
                }

                if(flag == 0) {
                    safe[count++] = i;

                    for(k=0; k<m; k++) {
                        avail[k] += alloc[i][k];
                    }

                    finish[i] = 1;
                }
            }
        }
    }

    printf("Safe Sequence:\n");

    for(i=0; i<n; i++) {
        printf("P%d ", safe[i]);
    }

    return 0;
}
```

```
}
```

Sample Input / Output

Sample Input:

Enter number of processes: 3

Enter number of resources: 3

Enter Allocation Matrix:

0 1 0

2 0 0

3 0 2

Enter Max Matrix:

7 5 3

3 2 2

9 0 2

Enter Available Resources:

3 3 2

Sample Output:

Safe Sequence:

P1 P2 P0